ZTP over UDP: Test Report

Version	1.2 (M7 release, after the final review and the `--bad-key` hook)
Owner	Sunil Vallamkonda
Code under test	commit `86d8ac2` (release regression)
Test plan	QA_TEST_PLAN.md: 165 cases (§5), impairment matrix (§6), traceability (§10)
Evidence	evidence/release-20261113-063341/ (the release run), evidence/ci/ (CI reports), logs/ (committed logs, logs/README.md)
Date	2026-11-13

This is the release test report that QA §11 asks for: the result of every test case, the impairment matrix with times and retransmission ratios, the performance numbers, and what is open or waived.

1. Summary

Result	Cases	Meaning
✅ PASS	144	Run and passed in the release regression (and in CI where noted)
⚠ Partial	6	Run and passed, except one part: either it needs the open test tooling (QA §12.5), or one platform or sub-case was not run (each row says which)
⚠ Open	14	Logic implemented, not exercised end to end: each needs a hostile-traffic tool (forged, replayed or flooded frames) that remains with the project owner (PRD R-15, QA §12.5)
⚠ Waived	1	TC-SIM-04 (macOS dummynet, needs `sudo`); TC-SIM-03 (netem in CI) and the in-code simulator cover the same behaviour
❌ Fail	0	
Total	**165**	

The demo runs the PDF's five devices and the committed dump has the PDF's five rows, including ZTP-84BC19 Failed / "Invalid HMAC" (a wrong model key, the `--bad-key` test hook; TC-AUTH-03, TC-PDF-02).

Runs behind these results

Run	Where	Result
`make release` on `86d8ac2`, 2026-11-13 06:33 → 09:13 PST	Development Mac (Darwin arm64, Apple clang 16.0.0), no sleep during the run	24/24 steps PASS; ASan build 16/16 (summary.txt)
CI on the final commits	GitHub Actions: Ubuntu gcc and clang, macOS clang, ASan on both, platform, matrix, netem, valgrind	all jobs green (README §17.2)
CI run 52 on `23757cd` (manual, nightly seed counts)	The same jobs, 10-seed matrix, TC-PERF-01/02, netem × 10	12/12 green (report); referenced below for the CI-only cases, whose code (transport, MTU, capture, netem paths) is unchanged since
Fresh clone of the final commit	Development Mac	`make`: no warnings; `make demo`: PASS (TC-DEL-01, TC-DEL-04)

2. Impairment matrix (QA §6)

The 5-device demo over each row, seeds 1–10, release run; the four valid devices are counted (ZTP-84BC19 must end FAILED, which each run's demo checks). "Retransmissions" is the share of all datagrams sent (server and devices, every run of the row) that were retransmissions; it and the out-of-order and duplicate counts are totals of the processes' `Counters:` lines (matrix_counters.txt has the table and the command; per-run checks and dumps are in `run_matrix/<row>.<seed>/`, and one run is kept whole in `run_matrix/M3.1/logs/`).

Row	Impairment (each direction)	Runs	Valid devices OPERATIONAL	Median	p95	Retransmissions	Out of order / dup received	Result
M1	none	10/10	40/40	0.4 s	0.4 s	0.0%	0 / 0	PASS
M2	10% loss, 5% reorder, 2% dup, 1 ms	10/10	40/40	1.4 s	3.4 s	11.9%	740 / 225	PASS
M3	15% loss, 10% reorder, 5% dup, 2±1 ms (the demo)	10/10	40/40	3.0 s	6.1 s	16.6%	1009 / 388	PASS
M4	20% loss, 10% reorder, 5% dup, 5±2 ms	10/10	40/40	4.8 s	12.2 s	20.5%	1065 / 405	PASS
M5	30% loss, 20% reorder, 10% dup (stress, ≥ 95% required)	10/10	40/40	12.0 s	38.3 s	28.1%	1051 / 610	PASS
M6	50% loss	not run						not part of the release run
M7	15% loss correlated 25%, 10% reorder, 5% dup, 5 ms (kernel netem, CI Ubuntu)	10/10 (CI run 52)	all					PASS
M8	20% loss device → server only	10/10	40/40	1.3 s	6.3 s	15.0%	28 / 293	PASS
M9	20% loss server → device only	10/10	40/40	1.6 s	4.6 s	13.6%	783 / 109	PASS

In CI (run 52) rows M1–M4 × 10 seeds passed on Ubuntu and macOS with every valid device OPERATIONAL. Every run also checks the invariants: no duplicate DB rows, no second REGISTER after a restart, and every artifact hash of an OPERATIONAL device verified (`tests/check_invariants.sh`). The PMTU black-hole detector re-requested chunks in 4 of the 360 device runs of the matrix and perf02 and never stepped down (DESIGN §5.8).

3. Performance (QA §8)

Case	Measure	Target	Result
TC-PERF-01	50 devices at 15% loss	all OPERATIONAL ≤ 120 s, server RSS ≤ 64 MB	50/50 in 24 s, peak RSS 9 MB (Mac); 17 s, 5 MB (CI macOS) and 24 s, 8 MB (CI Ubuntu), run 52
TC-PERF-02	Time to OPERATIONAL, row M4 × 20	median ≤ 60 s, p95 ≤ 90 s	median 3.9 s, p95 10.6 s (Mac); 3.7 s / 9.6 s (CI macOS) and 3.9 s / 10.5 s (CI Ubuntu), run 52

Case	Measure	Target	Result
TC-PERF-03	Transmissions per frame at 10 / 20 / 30% loss	within 10% of the model 1.23 / 1.56 / 2.04	1.16 / 1.35 / 1.68 (m6_perf03.out)
TC-PERF-04	1 h soak, 5 devices	no RSS growth, no leaks	every process's RSS lower at the end than after warm-up (server 1024 → 992 kB); `leaks` : 0 in all 6 processes; 9155 keepalives, 0 session timeouts; valgrind (CI, 3 min): clean
TC-NAT-07	30 min at 20% loss, production timers	0 session timeouts	0 timeouts, 87 keepalives, 1 REGISTER
TC-NAT-13	10 min at 20% loss, demo timers	timeouts recover without re-registration	1 timeout, recovered in 0.5 s, no re-registration

4. Defects found by the release runs and the final review

Found by	Defect	Fix
Release run on 3f9a06c, matrix M3 seed 3	Device: a false PMTU black-hole step-down to 508 B under ordinary loss	821d134: re-request once, step down only after a second silent interval (DESIGN §5.8)
Release run on 821d134	Device: full-size chunks buffered out of order were not counted as data	a1c5bd3: any authenticated CHUNK_DATA over 508 B restarts the interval
Release run on 10b6850, matrix M4 seed 8	Device: a step-down while the only missing chunk was a 262-byte last chunk (which no black hole drops)	86d8ac2: step down only if a missing chunk needs a datagram over 508 B
Release runs on 3f9a06c and a1c5bd3	Test harness: S4 rebinding check, TC-LOG-02 on loss-only rows, shared suite folders; the Mac slept during one run	821d134, b936db3 (the run fails itself if `pmset` logged a sleep)
CI on dc3c139	Test: TC-SRV-09 counted too few log lines to see interleaving	cfcb3fa
Final review (code)	Server: `limits.max_sessions` above 1536 could hang the session table; an Invalid HMAC REGISTER discarded the half-open session (a forged frame could abort a genuine registration); the rate limit was spent before the MAC check; NAK resends were unbounded; a device with a smaller datagram limit was stepped down to 508	0b2e460 (DESIGN §10.3, §5.8)
Final review (code)	DB journal: a torn last line could swallow the next record; non-ASCII strings made lines unreplayable; write/fsync errors were ignored before the ACK; the session history grew forever	0b2e460 (DESIGN §14.4)
Final review (code)	Device: the forced re-key at 2^{30} (DESIGN §8.9) was not implemented; a crash between the rename and the state save cost a hash attempt; a rollback could be reported as an upgrade after a crash; a queued probe counted as acknowledged	0b2e460
Final review (tests)	TC-UNIT-06 compared a constant with itself; weak checks in TC-UPG-02/03, TC-DISC-02, the `sha()` fallback and the GNU `nm` pattern	0b2e460
Final review (evidence)	The PDF dump's failed row was not demonstrated: no wrong-key device	10b6850: `--bad-key` ; TC-AUTH-03 end to end; the demo runs all five devices

One review finding is accepted as a residual risk: a replayed resume HELLO just after a server restart can stall a session briefly (DESIGN §18.3, item 5).

5. Results per test case

Each row: the case from QA §5, its result, where it ran, and the output that shows it. Suites marked "(Mac; CI build-test ×3, asan ×2)" are part of `make test`: they ran in the release run, in its ASan build, and in every CI run on Ubuntu (gcc, clang) and macOS, with and without AddressSanitizer. Each `.out` file is the suite's full output; a suite prints every failed check (`FAIL: TC-…`) and ends with its passed and failed counts, so a PASS row means none of that case's checks failed. To rerun one suite or case: README §16.3.

Unit (L1)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UNIT-01	SHA-256 known-answer tests	NFR-01, FR-18	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-02	HMAC-SHA256 known-answer tests	FR-13	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-03	Header and TLV encode/decode round trip	FR-05, BON-02	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-04	TLV parser bounds	NFR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-05	Constant-time compare	NFR-06	P2	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-06	Serial-number arithmetic and wraparound	FR-05	P1	✅ PASS	`unit`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	unit.out, m5_upgrade.out
TC-UNIT-07	RTO estimator, Karn, backoff	FR-06	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-08	Reorder buffer: in-order delivery	BON-04	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-09	Duplicate classification	FR-08, BON-05	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-10	Generating and parsing SACK	FR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-11	Policy parser and precedence	FR-29	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-12	Version comparison	FR-21	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-13	Atomic file write	NFR-10	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-14	Persisting and loading the chunk bitmap	FR-19	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-15	Netsim determinism and rates	NFR-09, BON-06	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-16	Key-derivation known-answer test	FR-13	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-17	Config key validation	FR-24	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UNIT-18	Strict JSON parser	FR-34, NFR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-19	Bitmap conversion on chunk-size change	FR-19, FR-20	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-20	Precedence and policy validation	FR-34, FR-35	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out

Discovery (DISC)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-DISC-01	Soft serial generated on first boot	FR-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m2_session.out, scenario_logs.out
TC-DISC-02	Soft serial loaded on later boots, and assigned by configuration	FR-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m2_session.out, scenario_logs.out
TC-DISC-03	Static IP discovery	FR-02	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-04	Multicast discovery	BON-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-05	Broadcast discovery	BON-01	P2	⚠ Partial	`m2_session` (Mac; CI build-test ×3, asan ×2); CI `build-test` (Ubuntu): broadcast, part (b)	m2_session.out, CI run 47 — part (b), broadcast on Ubuntu CI, runs; part (a), broadcast on the Mac, is skipped (no BCAST address set locally)
TC-DISC-06	Fallback to static	BON-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-07	Forged OFFER ignored	BON-01, NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-DISC-08	Replayed OFFER ignored	BON-01	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out

Session (SESS)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SESS-01	Handshake happy path	FR-04	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-02	HELLO lost: idempotent retransmission	FR-06, BON-05	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SESS-03	HELLO_ACK lost	FR-06, BON-05	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-04	Session IDs are random and unique	FR-04	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SESS-05	Server down: retry exhaustion and recovery	FR-06, FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-SESS-06	Unknown session_id dropped	NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SESS-07	Stale epoch dropped	FR-23	P1	⚠ Open	—	QA §12.5; needs an old-epoch frame injector
TC-SESS-08	UDP only	FR-03, DEL-04	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-09	A HELLO can't tear down a live session	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs a HELLO takeover tool

Reliability (REL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-REL-01	Every frame carries a seq	FR-05, BON-02	P1	✅ PASS	CI `platform-linux` : packet capture checked frame by frame	CI run 47
TC-REL-02	Retransmission on timeout	FR-06, FR-07, BON-03	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-REL-03	The PDF scenario, reproduced verbatim: seq 2 lost, seq 3 buffered, duplicate of seq 2	BON-04, BON-08	P1	✅ PASS	`unit` , `m2_peer` (Mac; CI build-test ×3, asan ×2)	unit.out, m2_peer.out
TC-REL-04	Lost ACK: data received	BON-07, BON-05	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-REL-05	Heavy duplication	FR-08, BON-05	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-REL-06	Window limit respected	FR-06	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-REL-07	Fast retransmit	FR-06	P2	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-REL-08	Out-of-window frames dropped	NFR-07	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-REL-09	Completes at 10/15/20% loss	FR-09, NFR-03	P1	✅ PASS	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, CI run 47
TC-REL-10	Heavy reordering	BON-04	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-REL-11	No ACK loops	FR-07	P2	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out

NAT and keepalive (NAT)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-NAT-01	Keepalives when idle	FR-10	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-02	No keepalives during a transfer	FR-10	P2	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-03	Device detects a dead server	FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m5_gate.out, scenario_logs.out
TC-NAT-04	Server detects an idle device	FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-05	NAT rebinding	FR-11, FR-23	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-NAT-06	Spoofed source can't hijack	NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-NAT-07	No false timeouts under loss (production timers)	FR-11, NFR-03	P2	✅ PASS	`longrun` (Mac)	longrun.out
TC-NAT-13	False timeouts with demo timers recover cleanly	FR-11, FR-23	P2	✅ PASS	`longrun` (Mac)	longrun.out
TC-NAT-08	NAT idle timeout shorter than the keepalive	FR-36, FR-35, FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-09	NAT reboot: all mappings flushed	FR-36, FR-23	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-10	Public IP change	FR-36	P2	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2); CI `build-test` (Ubuntu): strict-IP resume needs 127.0.0.2, which macOS lacks (the Mac run prints "skipped")	m5_upgrade.out, CI run 47

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-NAT-11	Outbound UDP blocked by a customer firewall	FR-36, A-09	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-NAT-12	A replayed valid frame from a new address can't move the peer	NFR-06, FR-14, FR-11	P1	⚠ Partial	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out — KEEPALIVE half passes; the CHUNK_REQ replay half needs a replay tool

Authentication and registration (AUTH)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-AUTH-01	REGISTER contents	FR-12	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-02	Valid registration	FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-03	Invalid HMAC (wrong model key)	FR-13, FR-30	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-04	Replay in the same session	FR-14	P1	⚠ Open	—	QA §12.5; needs `--replay-test`
TC-AUTH-05	Replay spliced into a new session	FR-14	P1	⚠ Open	—	QA §12.5; needs `--replay-test`
TC-AUTH-06	Reused nonce	FR-14	P1	⚠ Open	—	QA §12.5; needs `--reuse-nonce`
TC-AUTH-07	Replayed resume HELLO	FR-14, FR-23	P1	⚠ Open	—	QA §12.5; needs a resume replay tool
TC-AUTH-08	Tampering with any byte	NFR-06	P1	⚠ Open	—	QA §12.5; needs a bit-flip injector (TC-UNIT-03 flips every bit of a frame)
TC-AUTH-09	Unknown model	FR-29, FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-10	HELLO flood is rate-limited	NFR-05	P2	⚠ Open	—	QA §12.5; needs a HELLO flood tool
TC-AUTH-11	Rogue server rejected	NFR-06	P1	⚠ Open	—	QA §12.5; needs a rogue-server `peer_sim` mode
TC-AUTH-12	Known model, no policy	FR-29, FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-AUTH-13	Wrong fleet key (not a fleet device)	NFR-06, FR-36	P2	⚠ Open	—	QA §12.5; needs a wrong-fleet-key sender
TC-AUTH-14	Forged or replayed NAK is ignored	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs forged/replayed NAKs
TC-AUTH-15	Replayed terminal ERROR is ignored	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs replayed terminal errors

Provisioning (PROV)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PROV-01	Config in chunks with hash	FR-15, FR-17, FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-02	Config is valid JSON	FR-15	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-03	Firmware in chunks with hash	FR-16, FR-17, FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-04	Datagram size cap	FR-20, NFR-02	P1	✅ PASS	CI `platform-linux`: packet capture	CI run 47
TC-PROV-05	DF set, no IP fragments	FR-20	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2); CI `platform-linux`: packet capture	unit.out, CI run 47
TC-PROV-06	Chunk boundaries	FR-17	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-07	Hash mismatch handling	FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-08	Resume after kill -9	FR-19	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-09	Resume after a server restart	FR-19, FR-23	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PROV-10	Artifact changed between attempts	FR-18, FR-19	P2	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-11	Duplicate chunks are harmless	BON-05	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PROV-12	1280-byte MTU path: no fallback needed	FR-20, NFR-02	P1	✅ PASS	CI `platform-linux`: loopback MTU 1280	CI run 47
TC-PROV-13	Small-MTU path: step-down to 508 B	FR-20, FR-19	P1	⚠ Partial	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out — parts (b) and (c) run on the Mac and in CI; part (a), a real 576-byte MTU path, needs a CI job that was not built

Upgrade and restart (UPG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UPG-01	Outdated firmware upgraded	FR-21	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-02	Process restart	FR-22	P1	✅ PASS	`m5_gate`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_gate.out, m5_upgrade.out
TC-UPG-03	Resume without re-registration	FR-23	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-04	One DB row per serial	FR-23, FR-30	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-UPG-05	Already current: no upgrade	FR-21	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-06	Legacy device path	FR-29, FR-21	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-UPG-07	Crash at each upgrade step	NFR-10	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-08	Grace period expired before the resume	FR-23	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-09	Rollback on failed validation	FR-25	P2	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-10	Fast restart while the server session is still live	FR-23, FR-19	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-11	Crash during resume, then resume again (no deadlock)	FR-23, NFR-10	P1	✅ PASS	`m5_gate`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_gate.out, m5_upgrade.out

Validation (VAL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-VAL-01	Happy-path validation	FR-24..27	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-02	Config missing at validation	FR-24	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-03	Firmware version mismatch	FR-25	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-04	Unhealthy session	FR-26	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-VAL-05	Server records OPERATIONAL	FR-27, FR-30	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out

Server (SRV)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SRV-01	5-device concurrent demo	FR-28	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-02	Per-device assignment	FR-29	P1	✅ PASS	`unit`, `m6_server` (Mac; CI build-test ×3, asan ×2)	unit.out, m6_server.out
TC-SRV-03	DB record fields	FR-30	P1	✅ PASS	`unit`, `m3_register` (Mac; CI build-test ×3, asan ×2)	unit.out, m3_register.out
TC-SRV-04	Dump on demand	FR-31, DEL-11	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-05	Summary	FR-32	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-06	DB persists across a server restart	NFR-10	P1	✅ PASS	`unit`, `m6_demo` (Mac; CI build-test ×3, asan ×2)	unit.out, m6_demo.out
TC-SRV-07	Fuzzing malformed datagrams	NFR-07	P1	✅ PASS	`m6_fuzz` (Mac; CI build-test ×3, asan ×2)	m6_fuzz.out
TC-SRV-08	Session limits and half-open expiry	NFR-05, NFR-07	P2	⚠ Open	—	QA §12.5; needs a half-open flood tool
TC-SRV-09	Interleaved progress	FR-28	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-10	Resume across a server restart mid-upgrade	FR-23, NFR-10	P1	✅ PASS	`m6_server` (Mac; CI build-test ×3, asan ×2)	m6_server.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SRV-11	Failed-record cap	NFR-07, NFR-05	P2	⚠ Open	—	QA §12.5; needs 20,000 failing registrations (cap unit-tested at 1,000 over 100)

Simulation (SIM)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SIM-01	Observed loss rate, per direction and end to end	BON-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SIM-02	Reordering and duplication are observable	BON-06	P1	✅ PASS	`m2_session`, `m4_provision` (Mac; CI build-test ×3, asan ×2)	m2_session.out, m4_provision.out
TC-SIM-03	The netem alternative	BON-06	P2	✅ PASS	CI `impair-netem`	CI run 47
TC-SIM-04	The macOS dummynet alternative	BON-06	P2	⚠ Waived	—	Waived: macOS dummynet needs `sudo`; TC-SIM-03 and the in-code simulator cover it

Performance (PERF)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PERF-01	50 concurrent devices	NFR-05, FR-28	P1	✅ PASS	`scale` (Mac); CI `impair-matrix` (Ubuntu, macOS)	scale.out, CI run 47
TC-PERF-02	Time to OPERATIONAL	NFR-04	P1	✅ PASS	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, perf02.out, CI run 47
TC-PERF-03	Retransmission overhead against the model	FR-09	P3	✅ PASS	`m6_perf03` (Mac; CI build-test ×3, asan ×2)	m6_perf03.out
TC-PERF-04	Soak with no leaks	NFR-07	P2	✅ PASS	`soak` (Mac, 1 h); CI `valgrind`: 3 min under memcheck	soak.out, CI run 47

Logs (LOG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-LOG-01	Required tags present	FR-33	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-LOG-02	Bonus log lines present in the demo	BON-08	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-LOG-03	Format and files	NFR-08	P2	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-LOG-04	Counters at exit	NFR-08	P2	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out

Deliverables (DEL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-DEL-01	Clean build on both platforms	DEL-02, NFR-01, NFR-11	P1	✅ PASS	Fresh clone of the final commit, `make` (0 warnings), and the CI build jobs (Ubuntu gcc and clang, macOS clang; `-Werror`)	the binaries link only the C library: linkage_macos.txt
TC-DEL-02	The demo produces its artifacts	DEL-01, DEL-09, DEL-10, DEL-11	P1	✅ PASS	`make demo` (Mac, release run)	demo.out, logs/
TC-DEL-03	Device/server separation	DEL-03	P1	✅ PASS	`make check-separation` (suite `separation`)	separation.out
TC-DEL-04	The README quick start works	DEL-05	P1	✅ PASS	Fresh clone of the final commit, README quick start	`make` and `make demo` PASS
TC-DEL-05	README: 5 required answers	DEL-06	P1	✅ PASS	README §11 (review)	Q1–Q5 present
TC-DEL-06	README: design sections	DEL-07	P1	✅ PASS	README §4–§10 (review)	all 6 design topics present
TC-DEL-07	README: bonus coverage	DEL-08	P1	✅ PASS	README §13 (review)	3 topics and B1–B4 present
TC-DEL-08	Sanitizer-clean demo	NFR-11	P1	✅ PASS	`make asan` (ASan build, `m6_demo`)	asan/m6_demo.out
TC-DEL-09	Logs committed	DEL-09, DEL-10	P1	✅ PASS	`git ls-files logs/` ; `check_pdf05` (Mac; CI build-test ×3, asan ×2)	check_pdf05.out, logs/
TC-DEL-10	README: why keepalives matter	DEL-12	P1	✅ PASS	README §12 (review)	keepalive section present

End-to-end (E2E)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-E2E-01	Appendix A flow, single device	FR-33, FR-01..27	P1	✅ PASS	`check_appendix_a` (Mac and CI)	check_appendix_a.out
TC-E2E-02	Full demo under impairment	FR-28..32, NFR-03	P1	⚠ Partial	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, CI run 47 — all devices OPERATIONAL in every run; ZTP-71DD08 failed_attempts = 1 needs `--replay-test`

Conformance with PDF Appendix A (APX), one test per step

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-APX-01	Step 1: Device boot (device information: ZTP-93AF21, EDGE-GW-100, v1.0)	FR-01, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-02	Step 2: Server discovery (static IP 10.10.10.5, UDP port 5555)	FR-02, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2); CI `platform-linux` : on 10.10.10.5:5555	check_appendix_a.out, CI run 47
TC-APX-03	Step 3: UDP session (HELLO seq=1 with soft_serial; HELLO_ACK seq=1 with session_id=7421)	FR-03, FR-04, FR-05, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2); CI `platform-linux` : on 10.10.10.5:5555	check_appendix_a.out, CI run 47
TC-APX-04	Step 4: NAT keepalive	FR-10, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-05	Step 5: Secure registration (session_id=7421, soft_serial, model, firmware v1.0, nonce=984312)	FR-12, FR-13, FR-29, FR-33	P1	⚠ Partial	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out — REGISTER fields and order checked; the literal nonce 984312 needs `--nonce`
TC-APX-06	Step 6: Config provisioning (the example config, 3 chunks)	FR-15, FR-17, FR-18, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-07	Step 7: Firmware upgrade (v1.0 → v1.1)	FR-16, FR-21, FR-22, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-08	Step 8: Resume after restart, without duplicate registration	FR-22, FR-23, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-09	Step 9: Validation and operational mode	FR-24..27, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-10	Steps 1–9 under impairment	FR-09, NFR-03, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out

Conformance with the PDF's other examples and its scope (PDF)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PDF-01	§1 "Example output"	FR-01, FR-02, FR-33	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PDF-02	§9 "Example of a Dump from Server DB"	FR-31, FR-30, FR-32, DEL-11	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PDF-03	Appendix A "Example Configuration"	FR-15	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PDF-04	Appendix A addressing (10.10.10.5:5555)	FR-02	P3	✅ PASS	`check_appendix_a`, `m5_gate` (Mac; CI build-test ×3, asan ×2); CI `platform-linux`: on 10.10.10.5:5555	check_appendix_a.out, m5_gate.out, CI run 47
TC-PDF-05	Overview / Scope & Expectations: UDP only, "not a REST or API exercise", C or Python, no UI	FR-03, NFR-01, DEL-04	P1	✅ PASS	`check_pdf05` (Mac; CI build-test ×3, asan ×2)	check_pdf05.out

Configuration (CFG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-CFG-01	Built-in defaults	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-02	A config file overrides the defaults	FR-34	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-CFG-03	Precedence: defaults < file < profile < CLI	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-04	Range and type validation (fail fast)	FR-34, NFR-07	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-05	Unknown keys rejected (typo protection)	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-06	Per-tenant, model and serial NAT override delivered to the device	FR-35, FR-29	P1	✅ PASS	`m6_server` (the override in the server's policy line); `unit` (Mac; CI build-test ×3, asan ×2)	unit.out, m6_server.out
TC-CFG-07	Profiles	FR-34	P2	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-08	Device clamp on advertised values	FR-35, NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-CFG-09	Key-file permissions, and policy/artifact checks	FR-34, NFR-06	P1	✅ PASS	`unit`, `m3_register` (Mac; CI build-test ×3, asan ×2)	unit.out, m3_register.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-CFG-10	Settings from the resolved decisions take effect (DESIGN §23)	FR-34, FR-35	P2	⚠ Partial	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out — sub-case (f) runs (strict-IP resume, Linux CI); the other per-tenant/model/serial override sub-cases (a)-(e), (g) have no end-to-end check (the policy precedence is unit-tested, TC-UNIT-20)

6. Open and waived cases

The 14 open cases and the open parts of the partial ones that depend on tooling all need test tools that generate hostile or forged traffic: device hooks (`--replay-test`, `--reuse-nonce`, `--nonce`) and `peer_sim` modes (a rogue server, forged or replayed frames, floods). They were not built in this project and remain with the project owner (PRD R-15). They are counted as Open, not Waived: the owner accepted submitting without them, but they are not replaced by another test the way TC-SIM-04 is. The protection logic behind every one of them is implemented and was reviewed (DESIGN §18.3); README §18 lists, for each, which other tests reach that logic today. QA §12.5 is the register. The other partial cases are a platform or sub-case not run (each row says which).